

Online Simulation Versus Single-Point Evaluation for AKI Prediction: Identifying Optimal Prediction Horizons for Continuous Monitoring

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Abstract

Background: Acute kidney injury (AKI) affects up to 20% of hospitalized patients with significant mortality and costs. **Most AI prediction models are evaluated only at single time points, which may not reflect real-world continuous monitoring.**

Methods: An ITE Transformer model was developed using NHIS Ilsan Hospital data (n=83,085; 11.4% AKI). Three prediction horizons (0h, 48h, 72h) were evaluated via **online simulation across continuous time points** (0–72h before AKI).

Results: **0h model AUROC improved from 0.797 (72h before) to 0.933 (at event).** Sensitivity: 87.5%→92.2%, PPV: 42.4%→66.2%. The 72h model showed lower performance (AUROC 0.731–0.886).

Conclusions: **Online simulation reveals clinically important differences not apparent in single-point evaluation. The 0h model is optimal for continuous monitoring.**

Introduction

Acute kidney injury (AKI) is a common and serious complication of hospitalisation, affecting approximately 10–20% of hospitalised adults with substantial morbidity, mortality, and healthcare costs.

Deep learning models for AKI prediction have shown encouraging discrimination in retrospective studies. However, **model evaluation often does not reflect clinical practice.**

Most existing studies evaluate models at a single time point, yet clinical deployment involves **continuous monitoring** where models must generate reliable predictions repeatedly over time.

This evaluation gap may have important clinical implications. Models optimised for single-point performance could exhibit **less stable temporal behaviour during deployment.**

We developed and externally validated deep learning models for AKI prediction and propose **an online simulation framework** to evaluate performance under conditions approximating continuous clinical deployment.

Methods and Materials

Study Design and Data Sources

Retrospective cohort study using EHR data from three cohorts: NHIS Ilsan Hospital (development; n=83,085), Chuncheon Sacred Heart Hospital (external; n=68,615), and MIMIC-IV (external; n=5,623). Total: **157,323 admissions.**

Outcome and Features

AKI defined by KDIGO criteria (creatinine + urine output). **51 clinical features:** 3 demographics, 6 vital signs, 42 laboratory values.

Model Architecture

Initial Triplet Embedding (ITE) Transformer for irregularly-sampled clinical time series. Three prediction horizons: **0h, 48h, 72h** before AKI onset.

Online Simulation Framework

Predictions generated at **12-hour intervals from 72h before AKI to event time.** Models showing consistent AUROC improvement as AKI approached were considered more suitable for deployment.

Results

The 0h model demonstrated superior online simulation performance with AUROC improving from **0.797 (72h before AKI) to 0.933 (at event)**, gradient +0.136.

Sensitivity: **87.5%→92.2%**, PPV: **42.4%→66.2%** approaching event onset.

The 72h model showed consistently lower performance (AUROC 0.731–0.886) with **sensitivity at 65–67% near event time.**

48h model: intermediate (AUROC 0.812→0.916, gradient +0.104).

External validation:

- CSHH: 0h model AUROC **0.552→0.932**
- MIMIC-IV: 0h model AUROC **0.628→0.927**

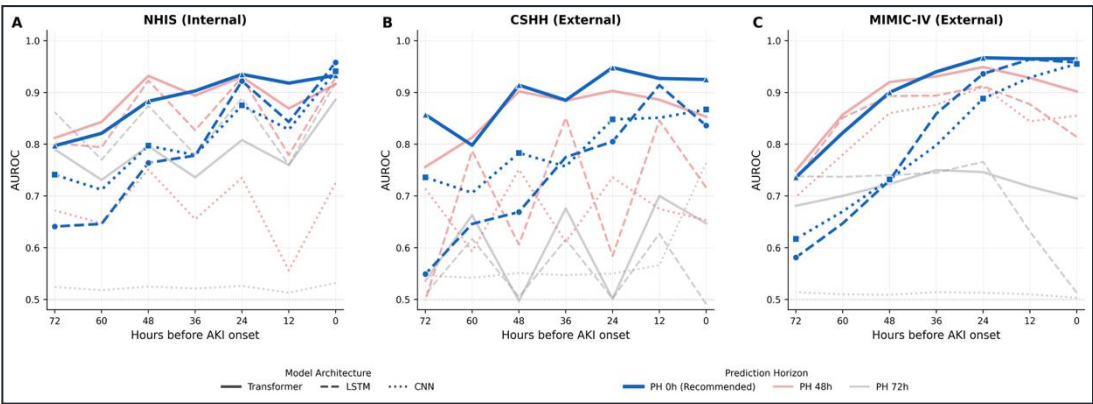


Figure 1. AUROC trajectories (72h → event) across 3 sites

	0h Model	48h Model	72h Model
AUROC (event)	0.933	0.916	0.886
AUROC (72h prior)	0.797	0.812	0.790
Gradient	+0.136	+0.104	+0.155
Sens. (event)	92.2%	92.5%	83.8%
PPV (event)	66.2%	63.1%	63.2%
Faithfulness	Strong ✓	Moderate △	Limited ✗

Discussion

- **Online simulation reveals clinically meaningful differences** between models that appear equivalent in single-point evaluation.
- **0h models are optimal for continuous monitoring** — accuracy increases as clinical deterioration progresses.
- Longer horizons face inherently **lower signal-to-noise ratio** before substantial renal dysfunction develops.
- **Shorter prediction horizons** may provide more effective early warning within continuous monitoring frameworks.
- **External validation confirmed generalisability** of temporal patterns.

Conclusions

1. **Online simulation reveals clinically important differences** not apparent in single-point evaluation.
2. **The 0h prediction horizon is optimal for continuous monitoring**, with adaptive behavior where accuracy increases as AKI approaches.
3. The 72h model showed **limited clinical utility** without meaningful improvement near event time.
4. External validation confirmed these patterns. **Temporal performance evaluation should be standard** for AI-based early warning systems.



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